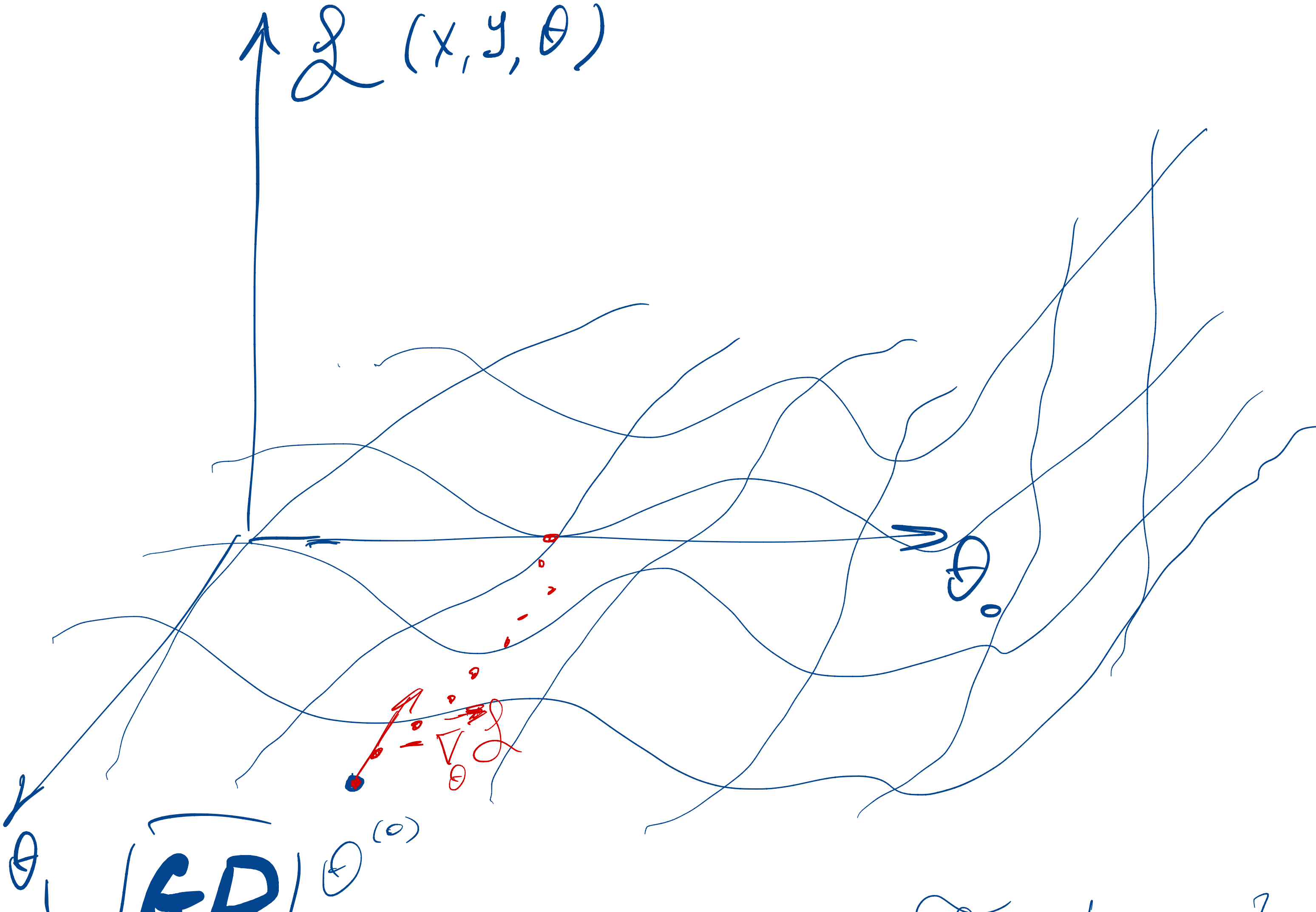


$\boxed{\textbf{GD}}$ $\theta^{(0)}$

$T = \{x, y\}$

1. $\vec{g}^{(t)} = \nabla_\theta \mathcal{L}(\theta^{(t)}, T)$
2. $\theta^{(t+1)} = \theta^{(t)} - \eta \vec{g}^{(t)}$
3. c? $\Rightarrow$ stop

SGD

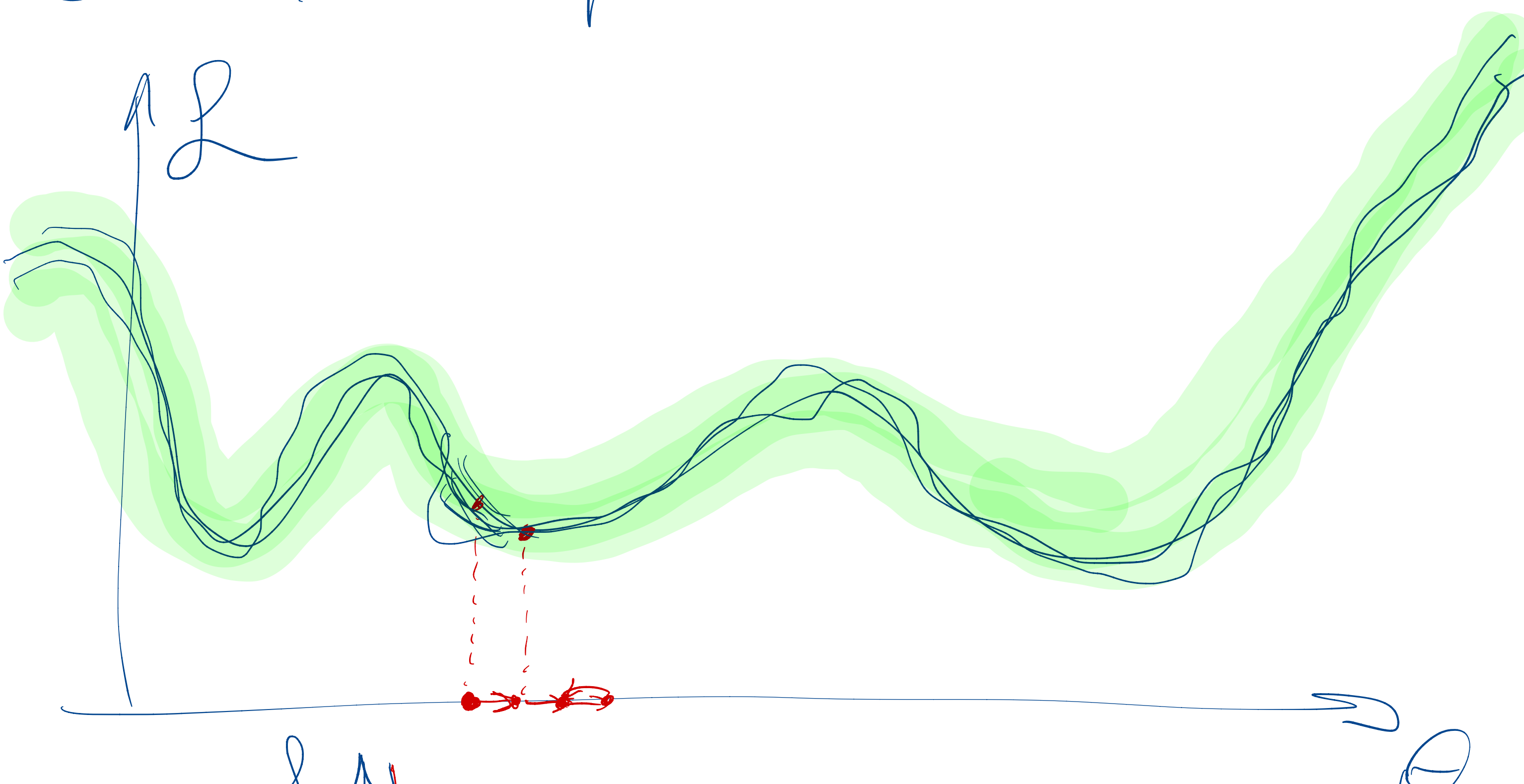
0. $\theta^{(0)}, \eta, |B|, C$

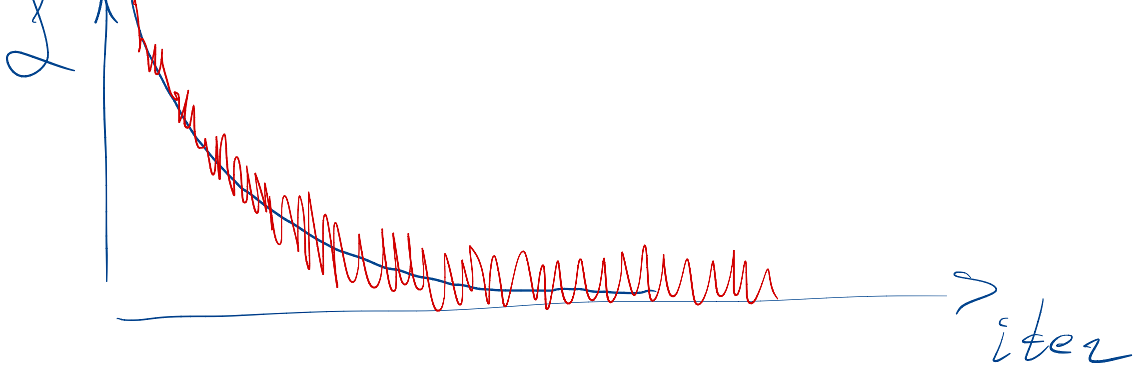
1. $\vec{g}^{(t)} = \vec{\nabla}_{\theta} \mathcal{L}(B, \theta^{(t)})$

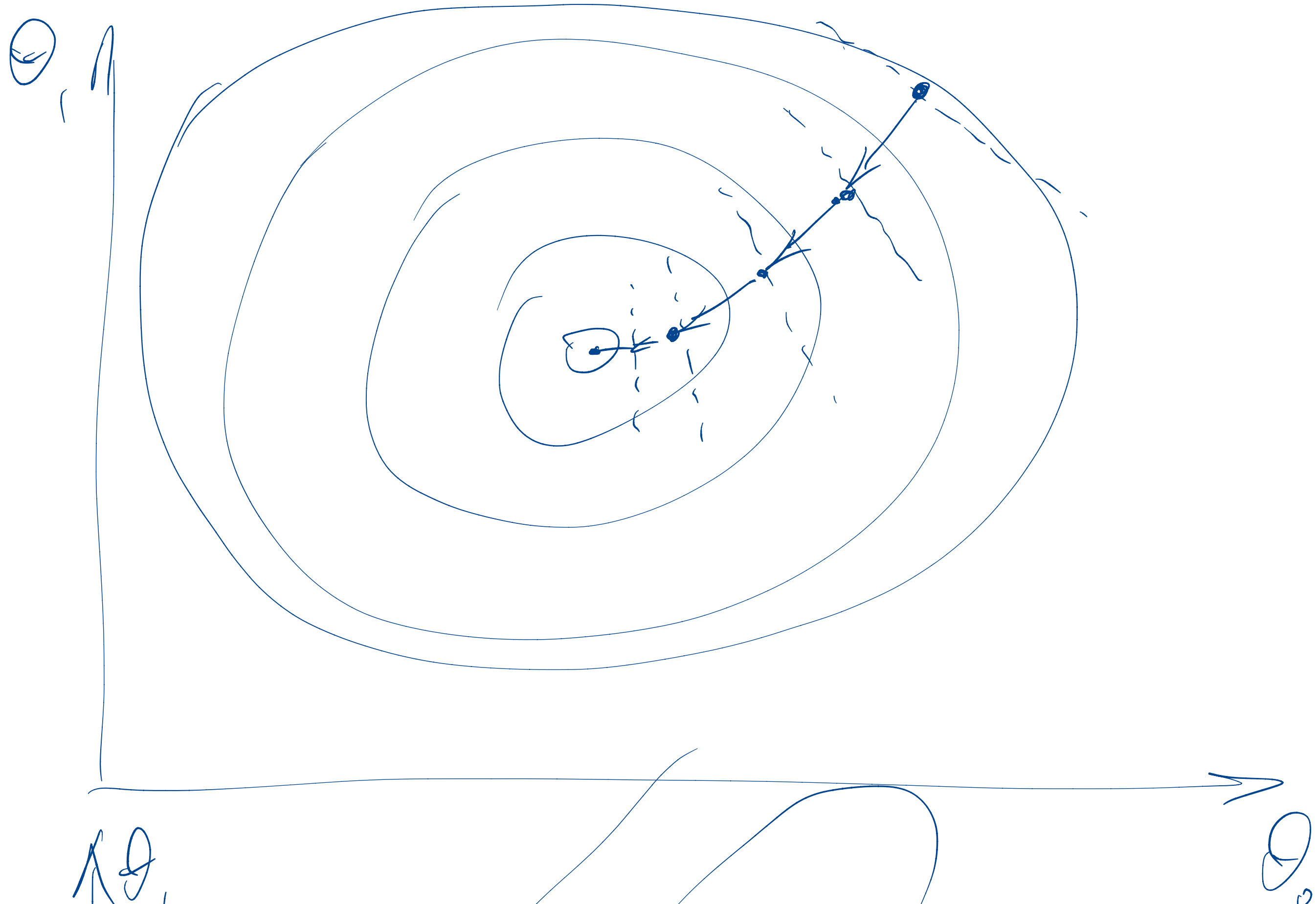
$B = \{x, y\}^{[bs]}$

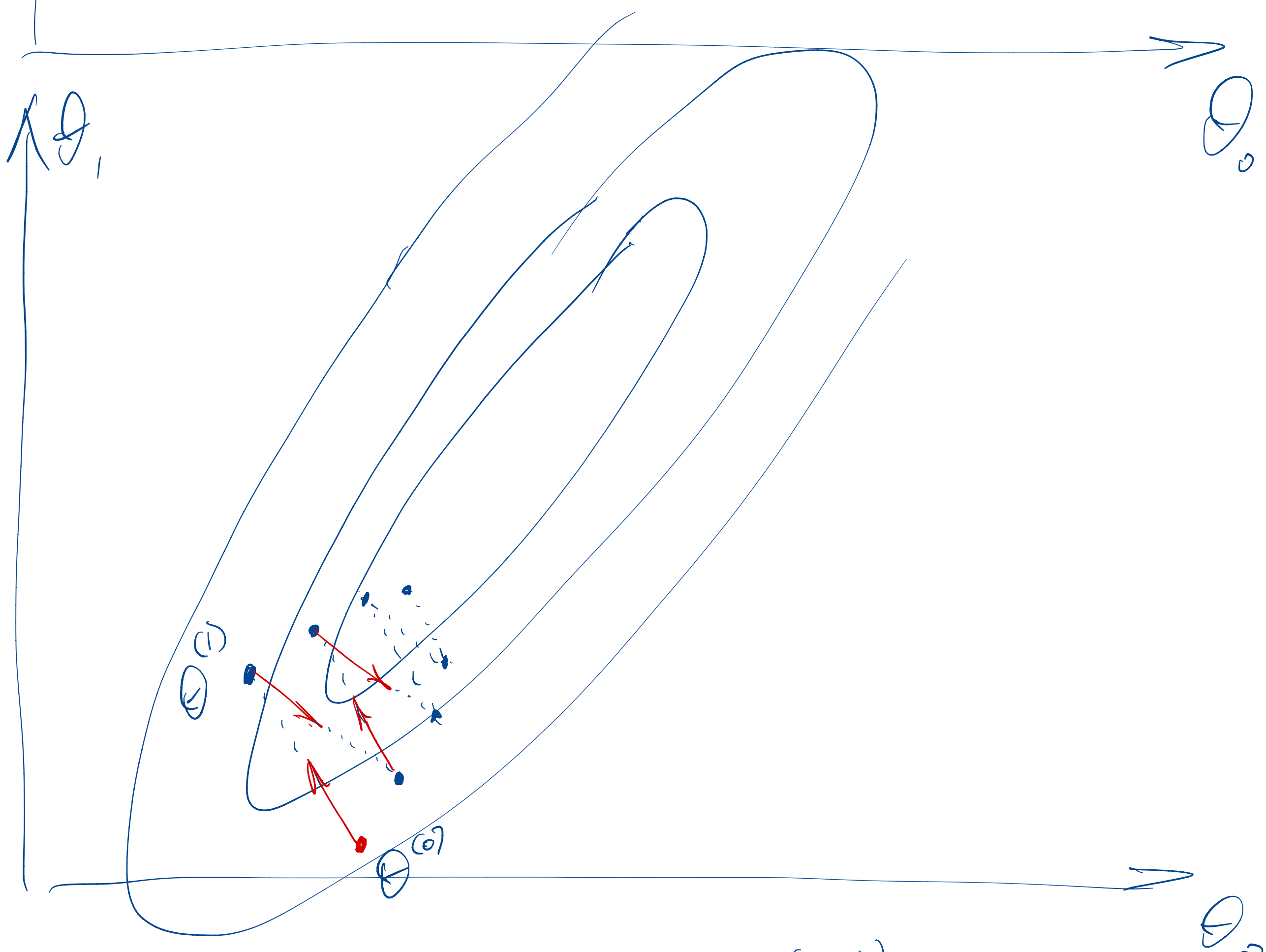
2. $\theta^{(t+1)} = \theta^{(t)} - \eta \vec{g}^{(t)}$

3. C? $\Rightarrow$ stop









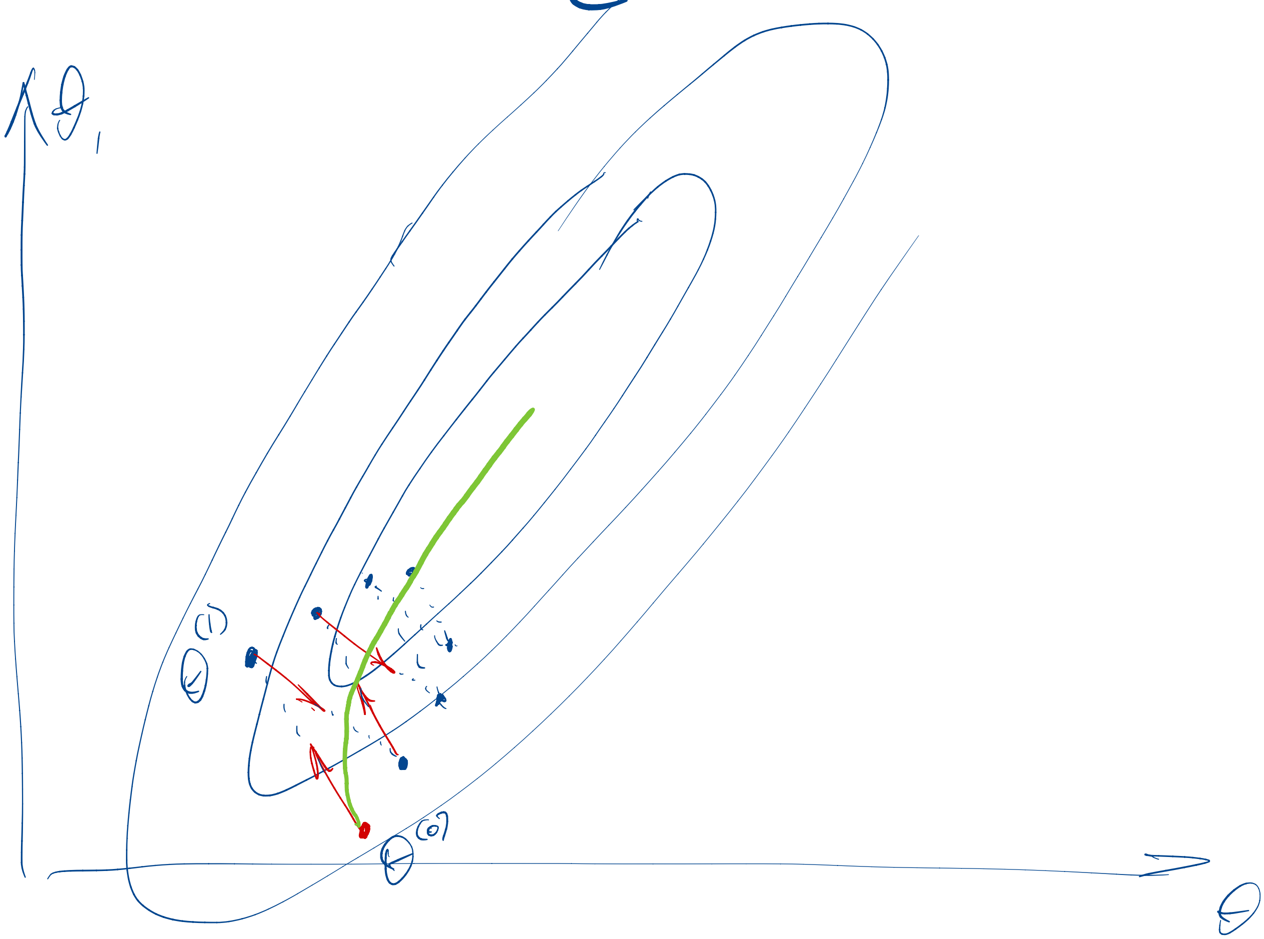
$$\vec{g} = \nabla_\theta \mathcal{L}(\theta^{(t)}, B) \qquad \theta^{(t+1)} = \theta^{(t)} - \eta g^{(t)}$$

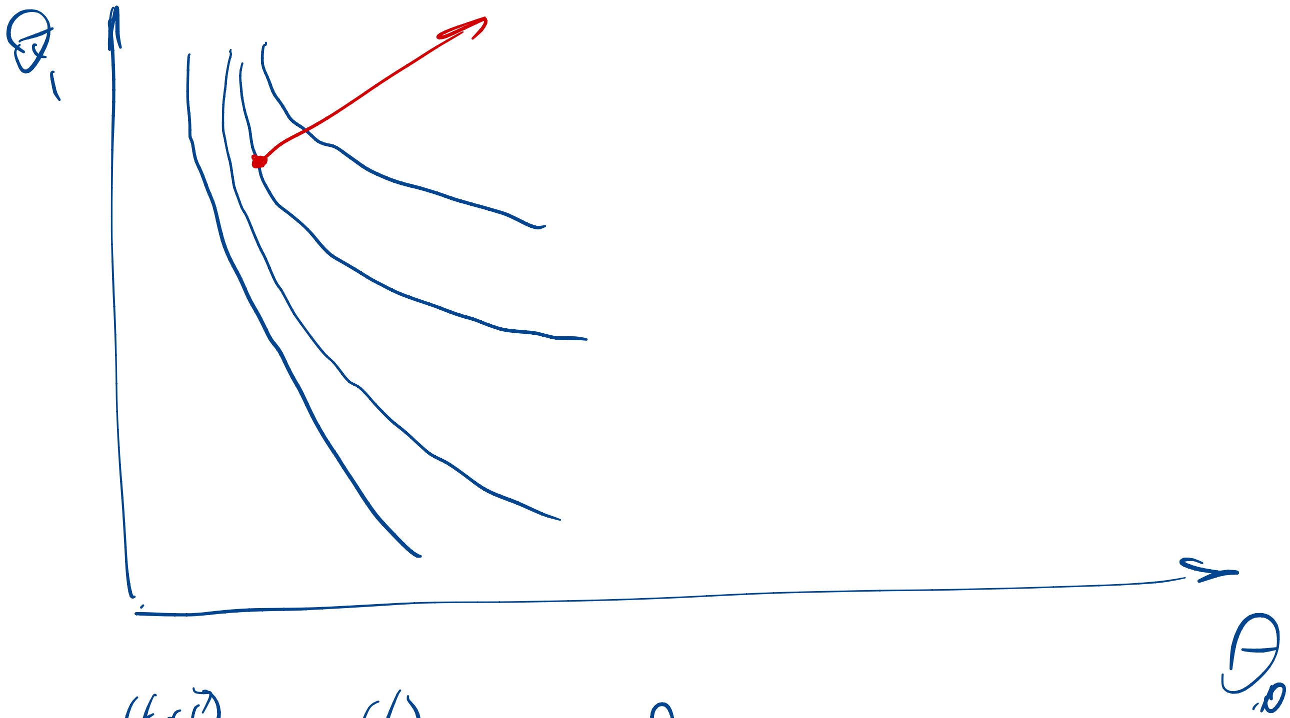
Импульсный метод
Momentum

$$\vec{g} = \nabla_\theta \mathcal{L}(\theta^{(t)}, B)$$

$$\vec{m}^{(t)} = \beta \vec{m}^{(t-1)} + (1-\beta)\vec{g}^{(t)}$$ Momentum

$$\theta^{(t+1)} = \theta^{(t)} - \eta \vec{m}^{(t)}$$





$$\Theta^{(t+1)} = \Theta^{(t)} - \eta \nabla_\Theta \mathcal{L}$$

$$\vec{m}^{(t)} = \beta \vec{m}^{(t-1)} + (1-\beta)\vec{g}^{(t)}$$

$$S = m^2$$

$$S_m = \beta_2 \left(S^{(t-1)}\right)^2 + (1-\beta_2)\left(m^{(t)}\right)^2$$

<u>RMSprop</u>

$$m' = \frac{m}{\sqrt{S_m + \epsilon}}$$

$$\Theta^{(t+1)} = \Theta^{(t)} - \eta m'$$

Adam: <u>Adaptive gradient + momentum</u>